

CHE 435/525 *Process Systems Analysis and Control*

Lecture 6: **Transfer Function and Local Linearization**

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Discuss & Recapture

- What is the inverse L.T. of $Y(s) = 1/(s - r)^k$?



- How to analyze the qualitative behavior of a signal based on its L.T.?
 - For example,

$$Y(s) = \frac{(s - 1)(s + 4)}{((s + 1)^2 + 2)(s + 3)}?$$

- What is a transfer function $G(s)$?

Learning objectives of this lecture

- ① Explain the meaning of transfer function and its **additive and multiplicative properties**
- ② Obtain the transfer function model from ODE using **deviation variables** around a steady state
- ③ Perform **local linearization** on nonlinear models around a steady state

Transfer function

Definition

For a linear SISO system with input $u(t)$ and output $y(t)$, the transfer function is

$$G(s) = Y(s)/U(s). \quad (1)$$



- An equivalent representation of the process dynamics
- A differential equation “disguised” as an algebraic equation

Transfer function: Example

Liquid storage tank with linear out-flow resistance

$$A \frac{dh}{dt} = q_i - \frac{1}{R} h.$$

Think of q_i (inlet flow rate) as the input, h as the output.

- By doing L.T.: $AsH(s) + (1/R)H(s) = Q_i(s)$

$$G(s) = \frac{H(s)}{Q_i(s)} = \frac{1}{As + 1/R}.$$

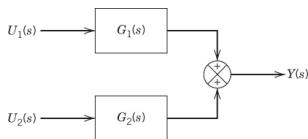
- Exercise: If $q_i(t) = M = \text{const}$ (step input), then how will $h(t)$ respond?

Additive and multiplicative properties

- If y is affected by multiple inputs (say, u_1 and u_2) with respective transfer functions ($G_1(s)$ and $G_2(s)$), then

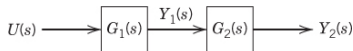
$$Y(s) = G_1(s)U_1(s) + G_2(s)U_2(s) \Rightarrow G(s) = G_1(s) + G_2(s)$$

For a linear system, the effect of inputs can be **superposed**.



- If y_2 is affected by input y_1 , which is the output of a precedent model $G_1(s)$ with input u . Then

$$Y_2(s) = G_2(s)Y_1(s) = G_2(s)G_1(s)U(s) \Rightarrow G(s) = G_2(s)G_1(s)$$



Static gain (steady-state gain, or gain)

- Under $u(t) = S(t)$ (unit step input), if $\lim_{t \rightarrow \infty} y(t) = K$ exists, then K is called the **static gain**.
 - ▶ If $u(t) = aS(t)$, then $\lim_{t \rightarrow \infty} y(t) = aK$

Formula of static gain

If static gain K exists, then

$$K = G(0).$$

- Proof: According to the final value theorem, $K = \lim_{s \rightarrow 0} sY(s)$, where $Y(s) = G(s)U(s)$, $U(s) = 1/s$. Hence $K = G(0)$.
- Question: What is the static gain of $G(s) = 1/(s + 3)$? What about $G(s) = 1/(s - 3)$?
- Question: What should be the static gain of a thermometer (input = object temperature, output = reading)?

Steady state and deviation variables

- A **steady state** refers to a status where the process variables remain at constant values
 - ▶ Assume model $\frac{dy}{dt} = f(y, u)$, the steady state is $(\bar{y}, \bar{u})$ such that $f(\bar{y}, \bar{u}) = 0$
 - ▶ Usually, linear dynamic models are only valid near a steady state
 - ▶ Think about a thermometer – only designed to measure near 36.5°C
- We should consider the **deviation from the steady-state values** as the inputs and outputs, not their “physical values”
 - ▶ Deviation variables: $y' = y - \bar{y}$, $u' = u - \bar{u}$
 - ▶ In other words, we study the response of y' to u'

Local linearization for $dy/dt = f(y, u)$

- R.H.S.: Near the steady state $(\bar{y}, \bar{u})$

$$\begin{aligned} f(y, u) &\approx f(\bar{y}, \bar{u}) + \left. \frac{\partial f}{\partial y} \right|_{(\bar{y}, \bar{u})} (y - \bar{y}) + \left. \frac{\partial f}{\partial u} \right|_{(\bar{y}, \bar{u})} (u - \bar{u}) \\ &= 0 + \left. \frac{\partial f}{\partial y} \right|_{(\bar{y}, \bar{u})} y' + \left. \frac{\partial f}{\partial u} \right|_{(\bar{y}, \bar{u})} u' \end{aligned}$$

- L.H.S.: $dy/dt = d(\bar{y} + y')/dt = dy'/dt$.

- Denote

$$a = \left. \frac{\partial f}{\partial y} \right|_{(\bar{y}, \bar{u})}, \quad b = \left. \frac{\partial f}{\partial u} \right|_{(\bar{y}, \bar{u})},$$

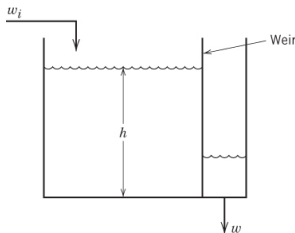
we have

$$\frac{dy'}{dt} = ay' + bu' \quad \Rightarrow \quad G(s) = \frac{Y'(s)}{U'(s)} = \frac{b}{s - a}.$$

Local linearization for $dy/dt = f(y, u)$: Example

Tank with weir [Seborg Ex.4.8]

A surge tank is designed with a slotted weir so that the outlet flow rate $w = Rh^{1.5}$. Find the transfer function $G(s) = H'(s)/W_i'(s)$.



$$\rho A \frac{dh}{dt} = w_i - Rh^{1.5}.$$

- Assume the steady state: $(\bar{h}, \bar{w}_i)$ satisfying $\bar{w}_i - R\bar{h}^{1.5} = 0$.
- Linearize: $\rho A \frac{dh'}{dt} = w_i' - 1.5R\bar{h}^{0.5}h'$. Hence, $G(s) = \frac{1}{\rho As + 1.5R\bar{h}^{0.5}}$.

Local linearization: Multi-input-multi-output case

CSTR [Seborg Ex.4.10]

A CSTR with first-order reaction $A \rightarrow B$ using cooling water:

$$V \frac{dc_A}{dt} = q(c_{Ai} - c_A) - k_0 e^{-E/RT} c_A V$$

$$\rho V C_p \frac{dT}{dt} = \rho q C_p (T_i - T) + k_0 e^{-E/RT} c_A V (-\Delta H_R) + UA(T_c - T)$$

Assume $c_{Ai} = \text{const.}$ Find the transfer function $C'_A(s)/T'_c(s)$.

- Assume the steady state: $(\bar{c}_A, \bar{T})$ under $\bar{T}_c$, making R.H.S. both zero.
- Linearize: $\frac{dc'_A}{dt} = a_{11}c'_A + a_{12}T'$, $\frac{dT'}{dt} = a_{21}c'_A + a_{22}T' + bT'_c$.
- L.T.: $(s - a_{11})C'_A(s) = a_{12}T'(s)$, $(s - a_{22})T'(s) = a_{21}C'_A(s) + bT'_c(s)$.
- Eliminate T :

$$\frac{C'_A(s)}{T'_c(s)} = \frac{a_{12}b}{s^2 - (a_{11} + a_{22})s + (a_{11}a_{22} - a_{12}a_{21})}.$$

Summary: Laplace transform and transfer function

- L.T.: signal in time, $f(t) \rightarrow$ signal in “ s ”, $F(s)$
 - ▶ Later, the “physical” meaning of s shall become clearer.
- Input-output model: differential in $t \rightarrow$ algebraic in s

$$G(s) = Y(s)/U(s)$$

- ▶ If nonlinear, then locally linearize near a steady state
- Calculating output response:

$$u(t) \xrightarrow{\text{L.T.}} U(s) \xrightarrow{\text{T.F.}} Y(s) \xrightarrow{\text{P.F.E.}} y(t)$$

- **Qualitative analysis** possible! Examine the roots on the denominator
 - **real part (growth/decay) and imaginary part (oscillations)**
 - ▶ Next few lectures: Apply such analysis to systems with different “features”